Ultimate Nutrition

Premium Inosine

Energy is what the muscles of athletes involved in aerobic, high-intensity exercise require. This short-duration energy requirement is produced by the breakdown of glycogen, which produces ATP (adenosine triphosphate), the currency of bodily energy. This is also called the aerobic pathway of energy production. After two to three minutes of intense exercise, aerobic pathway becomes the major source of energy production as the body begins to use oxygen to burn nutrients for energy. In aerobic metabolism, the body uses glucose, fatty acids and amino acids to produce energy that the muscles rely upon.

Inosine is the perfect training companion. It is a natural substance that can help increase the body's ability to carry oxygen, without affecting its uptake in the lungs, so it can work harder, stronger. When maximum stimulation is reached Inosine is converted into nutrients used by the body.

Ultimate Nutrition's Premium Inosine supports the body in production of short bursts of energy during intense, aerobic exercise. Importantly, inosine increases the body's ability to handle rigorous exercise without inducing fatigue. What is Inosine? Inosine, also known as hypoxanthine riboside, belongs to a class of naturally occurring compounds called purine nucleotides, which are the building blocks of DNA and RNA. Adenosine in ATP is also a purine nucleotide.

Inosine penetrates the cell walls of both the heart and skeletal muscle and, once inside the cell, promotes the production of ATP, which helps the muscle to contract. In addition, inosine also facilitates the transport of oxygen from the circulating blood to various tissues in the body including the muscle. Combination of these two important properties makes it a valuable dietary supplement for athletes whose goal is to exercise for longer periods of time without experiencing muscle fatigue.

Inosine is also known to activate various enzyme systems in the body, and it plays a crucial role in protein synthesis. Availability of abundant amounts of the protein in the muscle is necessary in order for the muscle to repair and grow, remodeling the tissue that the athletes need for performance. Furthermore, in aerobic energy production, protein is first broken down into amino acids before it can be used as a fuel to produce energy. Therefore, nutritional supplementation with inosine for enhanced performance during high-intensity aerobic exercise can hardly be overemphasized.

Ultimate Nutrition's Premium Inosine guarantees the highest potency and maximal effectiveness in fulfilling your energy needs. Being of the highest quality, Premium Inosine not only provides a ready source of energy production but also supports faster muscle recovery and, simultaneously, helps the body to tone and remodel the muscle for enhanced performance.

SELECTED REFERENCES:

1. Starling et al., "Effect of Inosine Supplementation on Aerobic and Anaerobic Cycling Performance," Med. Sci. Sports Exer.: 28, 1193, 1996
2. Rosenbloom et al., "Contemporary Ergogenic Aids Used by Strength/Power Athletes," J. Am. Diet Assoc.: 92, 1264, 1992
3. DiPrampero, P., "Energetics of Muscular Exercise," Biochem. Pharmacol.: 89, 143, 1981
4. Jacobs et al., "Sprint Training Effects on Muscle Myoglobin, Enzymes, Fiber Types and Blood Lactate," Med. Sci. Sports Exer.: 19, 369, 1987
5. Petersen, A. & Quistorff, B., "Inosine/Pyruvate/Phosphate Medium but not Adenosine/Pyruvate/Phosphate Medium Introduces Millimolar Amounts of 5-Phosphoribosyl 1-Pyrophosphate in Human Erythrocytes: A 31P-NMR Study," Biochem. J.: 266, 441, 1990
6. Kovacevic et al., "Interaction of Metabolism of Aspartate and Inosine and Energy State of Human Cells," Biochem. J.: 247, 47, 1987
7. Brouns, F., Nutritional Needs of the Athlete, Chichester, U. K.: John Wiley & Sons, 1993

FREQUENTLY ASKED QUESTIONS:

1. **In What Natural Sources in Inosine Found?**
   Inosine is more abundantly found in Brewer's yeast and in edible organ meats.
2. **Does Inosine Have Any Health Benefits Beyond Athletic Sports?**
   The major role of inosine in the cell is to produce energy and to help transport oxygen to various tissues. Even though inosine is presently used solely to provide bursts of energy in aerobic, active research is being carried out to discover any other potential applications of this invaluable nutritional supplement.
3. **If Inosine and Adenosine Are Similar, Why Not Use Adenosine?**
   While inosine and adenosine are chemically similar and belong to the same family of compounds, research has shown that nutritional supplementation with inosine, and not adenosine, offers maximum benefits to high-intensity athletes.
4. **What is the Recommended daily Dosage of Inosine?**
   Athletes tend to take rather large dosages of inosine, which can vary between 2000 milligrams to 5000 milligrams. To derive maximal benefits, dosages should be tailored to individual needs and tolerance levels. The help of a physical trainer and/or healthcare professional may also be necessary to determine optimal inosine dosage.
5. **What Are the Side Effects?**
   No side effects associated with the ingestion of inosine have been reported to-date. Supplemental inosine is contraindicated in those with a history of gouty arthritis with acute attacks.